Core2

SKU:K010















Description

Core2 is the second generation main unit in the M5Stack development kit series, further enhancing the functionality of the original first-generation main unit with more comprehensive hardware features. The core controller **Core2** is equipped with **ESP32-D0WDQ6-V3**, featuring two independently controllable **Xtensa® 32-bit LX6** processors with a main frequency of up to 240 MHz, supporting Wi-Fi

functionality. It has onboard **16MB Flash** and **8MB PSRAM**, and programs can be downloaded via the **USB Type-C** interface. Its powerful configuration meets the resource demands of complex applications. The front is equipped with a 2.0-inch integrated capacitive touch screen, providing users with a smoother human-computer interaction experience.

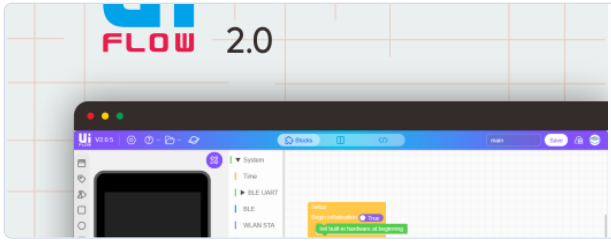
The body has a built-in **vibration motor**, providing **haptic feedback** and vibration alert functions. The built-in RTC module provides precise timing functionality. The power section is equipped with an **AXP192** power management chip that effectively controls the power consumption of the body, with a built-in green power indicator light. The body also includes a microSD slot and a speaker. To ensure higher quality sound effects, it uses a **power amplifier chip** with an **I2S** digital audio interface, effectively preventing signal distortion. The left side and bottom of the body have independent power and reset (RST) buttons, and the three dots on the front of the screen are part of the touch screen, which can be programmed to map hot zones as three virtual buttons. The back of the body integrates a small expansion board with a 6-axis IMU sensor and a microphone.

Tutorial



UIFlow

This tutorial introduces how to control the Core2 device through the UIFlow graphical programming platform.



UiFlow2

This tutorial introduces how to control the Core2 device through the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial introduces how to program and control the Core2 device using the Arduino IDE.

Note

- The built-in vibration motor of Core2 interferes structurally with the Base series bases. To prevent equipment damage, do not stack Core2 with any Base series functional bases.
- When stacking Core2 with M5 modules, you need to remove the battery bottom of Core2. If you want to retain the I2S microphone, IMU, and battery functions of the base while stacking other modules, the M5GO Bottom2 is recommended. The PCB of Core2 has reserved interfaces for the CP2104 chip and a lithium battery connector.
- Some screens may exhibit non-linear touch response at the edges. You can try upgrading the screen firmware using M5Tool to resolve this issue.

Features

- Developed based on ESP32, supports WiFi
- 16MB Flash, 8MB PSRAM
- Built-in speaker, power indicator light, vibration motor, RTC, I2S amplifier, capacitive touch screen, power button, reset button
- TF card slot
- Built-in lithium battery with power management chip
- Independent small board with built-in 6-axis IMU, PDM microphone
- M5-Bus socket
- Development Platform

- UiFlow1
- UiFlow2
- Arduino IDE
- ESP-IDF
- PlatformIO

Includes

- 1 x Core2
- 1 x USB Type-C cable (20cm)
- 1 x Hex wrench

Applications

- IoT controller
- STEM education
- DIY projects
- Smart home devices

Specifications

Specification	Parameter
SoC	ESP32-D0WDQ6-V3 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency 240MHz
DMIPS	600
SRAM	520KB
Flash	16MB
PSRAM	8MB Quad
Wi-Fi	2.4 GHz Wi-Fi
Input Voltage	5V@500mA
Host Interface	USB Type-C x 1, GROVE (I2C+I/O+UART) x 1
LED	Green power indicator LED
Button	Power button, RST button, screen virtual buttons x 3
Vibration Alert	Vibration motor
IPS LCD Screen	2.0"@320 x 240 ILI9342C / ILI9342E
Capacitive Touch IC	FT6336U
Speaker Amplifier	NS4168
Microphone	SPM1423
I2S Amplifier	NS4168
IMU	MPU6886
RTC	BM8563
PMU	AXP192
USB Chip	CP2104/CH9102F (two chip versions, no functional or usage differences)

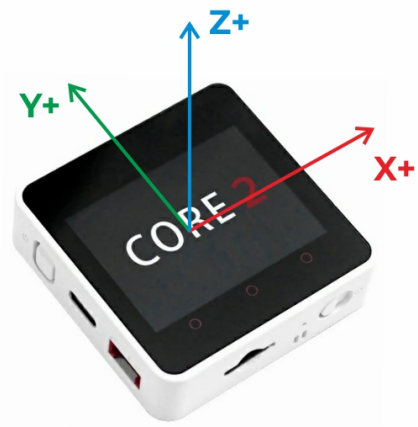
Specification	Parameter
DC-DC Boost	SY7088
Lithium Battery	500mAh@3.7V
Charging Parameters	Charging current: 0.219A Fully charged current (powered off): 0.055A Fully charged current (powered on): 0.147A
Antenna	2.4G 3D antenna
Operating Temperature	0 ~ 60°C
Base Screw Spec	Hex socket countersunk head M3
Reserved Internal PCB Ports	Battery port (Spec: 1.25mm-2P) USB cable port (Spec: 1.25mm-4P)
Product Size	54.0 x 54.0 x 16.5mm
Product Weight	54.9g
Package Size	80.0 x 59.9 x 21.6mm
Gross Weight	100.8g

Learn

Power On/Off

- Power on: Single-click the left power button
- Power off: Press and hold the left power button for 6 seconds
- Reset: Single-click the RST button on the bottom

IMU Triaxial Direction Schematic Diagram



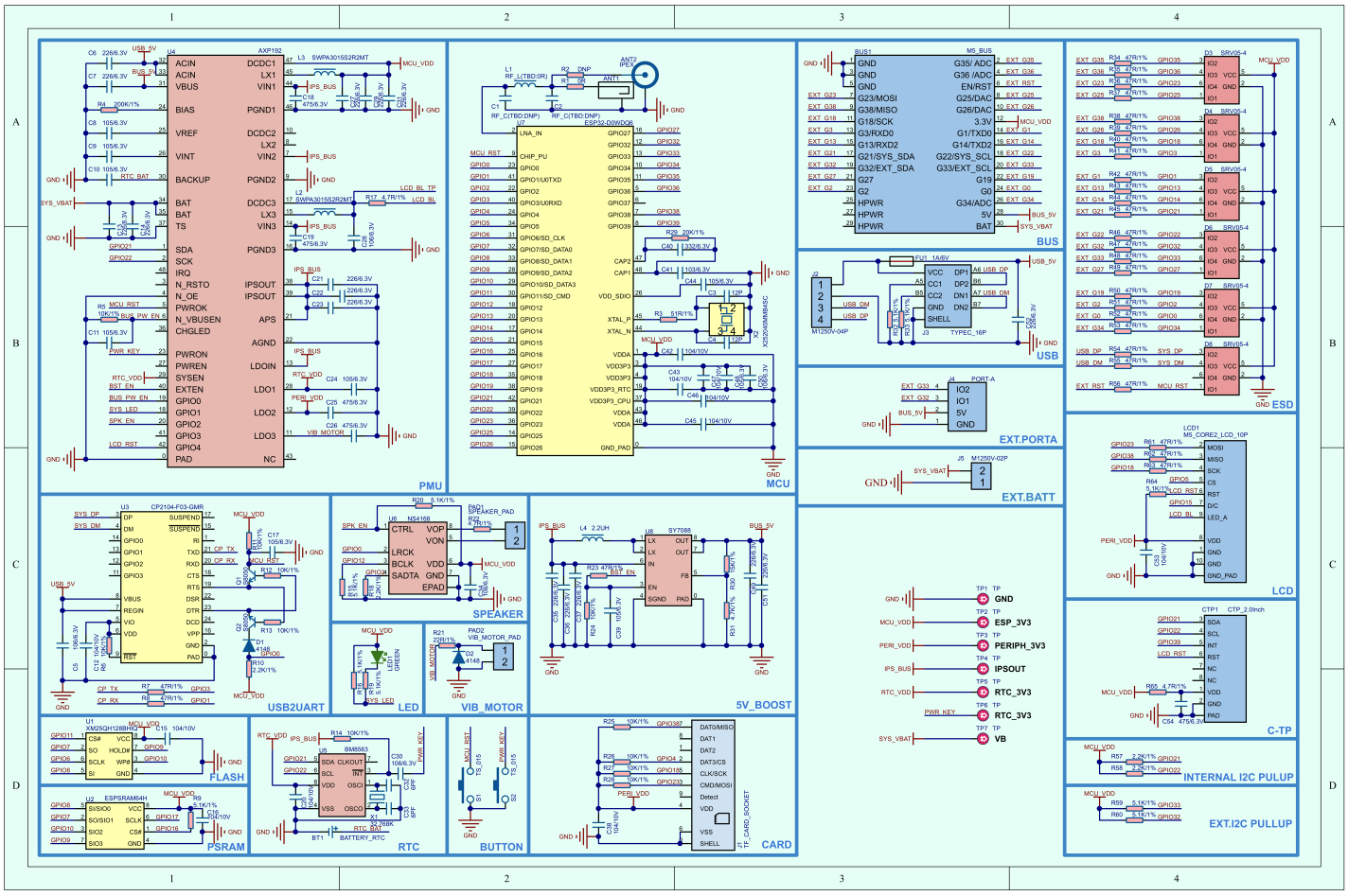
IMU ORIENTATION OF AXES

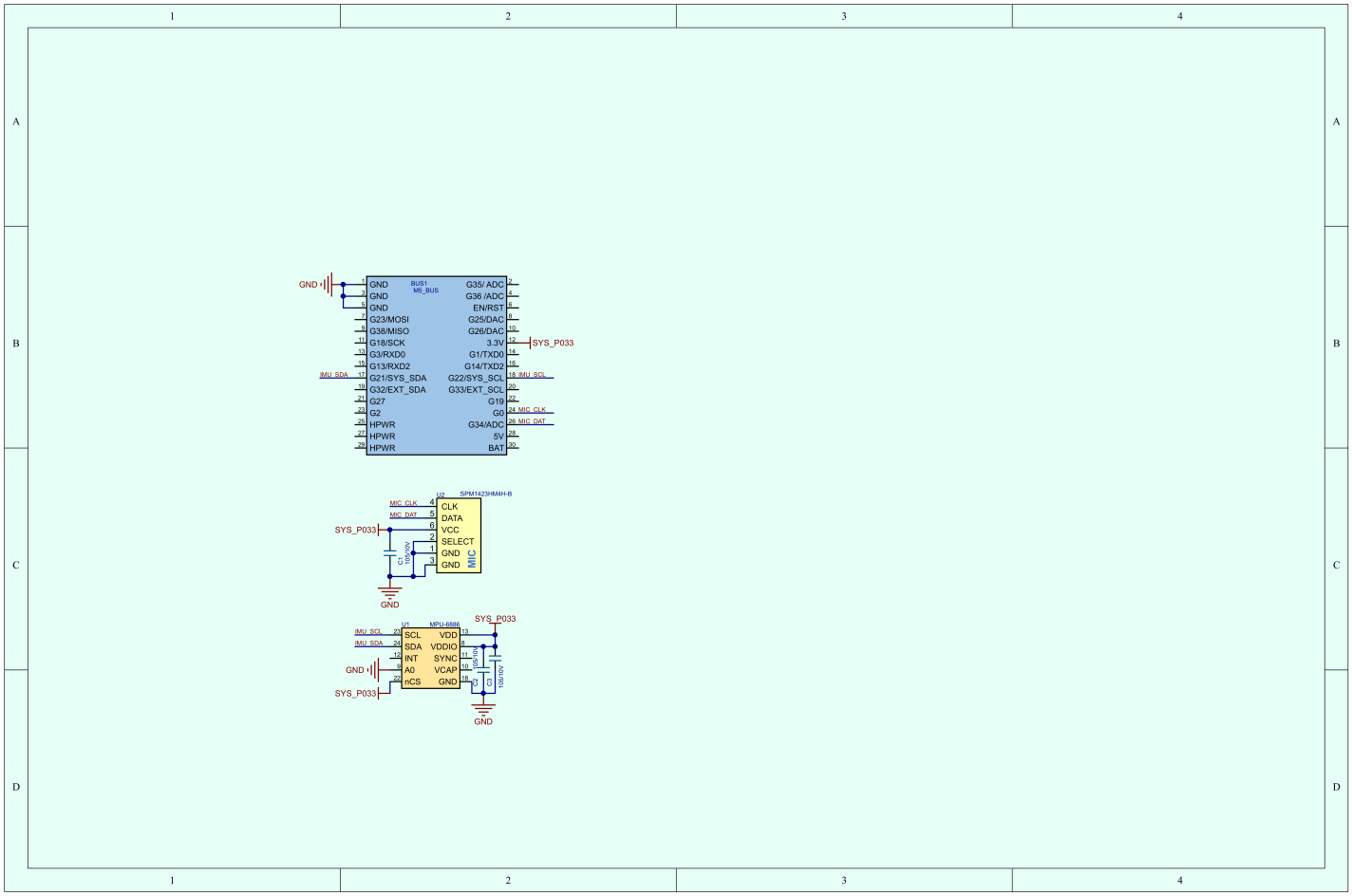
Certifications

- CE/MIC/FCC/RCM
- IEC62133

Schematics

- Core2 - Core Section Schematics PDF
- Core2 - Expansion Board Schematics PDF





PinMap

LCD Screen & TF Card

LCD Pixels: 320x240

ESP32-D0WDQ6-V3	G38	G23	G18	G5	G15
ILI9342C / ILI9342E	MISO	MOSI	SCK	CS	DC

AXP192	AXP_IO4	AXP_DC3	AXP_LDO2
ILI9342C / ILI9342E	RST	BL	PWR

microSD

ESP32-D0WDQ6-V3	G38	G23	G18	G4
TF Card	MISO	MOSI	SCK	CS

CAP.TOUCH

ESP32-D0WDQ6-V3	G21	G22	G39
FT6336U (0x38)	SDA	SCL	INT

AXP192	AXP_IO4
FT6336U	RST

Microphone & NS4168 Amplifier

ESP32-D0WDQ6-V3	G12	G0	G2	G34
NS4168	BCLK	LRCK	DATA	
Mic		CLK		DATA

AXP192	AXP_IO2
NS4168	SPK_EN

AXP Power Indicator & Vibration Motor

AXP192	AXP_IO1	AXP_LDO3
Green LED	Vcc	
Vibration Motor		Vcc

RTC

ESP32-D0WDQ6-V3	G21	G22
BM8563 (0x51)	SDA	SCL

AXP192	AXP_PWR
BM8563	INT

IMU (3-axis Gyroscope + 3-axis Accelerometer)

ESP32-D0WDQ6-V3	G21	G22
MPU6886 (0x68)	SDA	SCL

USB to Serial Download

ESP32-D0WDQ6-V3	G1	G3
CP2104/CH9102F	RXD	TXD

Internal I2C Connection

ESP32-D0WDQ6-V3	G21	G22
MPU6886	SDA	SCL
AXP192 (0x34)	SDA	SCL
BM8563	SDA	SCL
FT6336U	SDA	SCL

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G32	G33
PORT.B	GND	5V	G26	G36
PORT.C	GND	5V	G14	G13

M5-Bus

FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	G35	ADC
	GND	3	4	G36	ADC
	GND	5	6	RST	EN
MOSI	G23	7	8	G25	DAC
MISO	G38	9	10	G26	DAC
SCK	G18	11	12	3V3	
RXD0	G3	13	14	G1	TXD0
RXD2	G13	15	16	G14	TXD2
Int SDA	G21	17	18	G22	Int SCL
PORT.A SDA	G32	19	20	G33	PORT.A SCL
GPIO	G27	21	22	G19	GPIO
I2S_DOUT	G2	23	24	G0	I2S_LRCK
	NC	25	26	G34	I2S_DATA
	NC	27	28	5V	
	NC	29	30	BAT	

M5Core2 BUS (Comparison with M5Stack)

M5Stack	
LCD	
GPIO23	MOSI
GPIO18	SCK
GPIO14	CS
GPIO27	R/S
GPIO33	RST#
GPIO32	Backlight
Button	
GPIO37	Btn C
GPIO38	Btn B
GPIO39	Btn A
SDCard	
GPIO23	SDDI
GPIO19	SDDO
GPIO18	SDCLK
GPIO4	SDCS
Speaker	
GPIO25	INP

M5Core2 Bus										
M5Stack Bus										
GND			GND	1	2	ADC1	GPIO35	GPIO35	ADC
GND			GND	3	4	ADC2	GPIO36	GPIO36	ADC
GND			GND	5	6	RESET	EN	EN	RST
MOSI	GPIO23	GPIO23	MOSI	7	8	AUDIO_L	GPIO25	GPIO25	DAC
MISO	GPIO38	GPIO19	MISO	9	10	AUDIO_R	GPIO26	GPIO26	DAC
SCK	GPIO18	GPIO18	SCK	11	12	3.3V			3.3V
RXD0	GPIO3	GPIO3	RXD1	13	14	TXD1	GPIO1	GPIO1	TXD0
RXD2	GPIO13	GPIO16	RXD2	15	16	TXD2	GPIO17	GPIO14	TXD2
intSDA	GPIO21	GPIO21	SDA	17	18	SCL	GPIO22	GPIO22	intSCL
PA_SDA	GPIO32	GPIO2	IO6	19	20	IO7	GPIO5	GPIO33	PA_SCL
GPIO	GPIO27	GPIO12	IIS_SCLK	21	22	IIS_WS	GPIO13	GPIO19	GPIO
I2S_OUT	GPIO2	GPIO15	IIS_OUT	23	24	IIS_MCLK	GPIO0	GPIO0	2S_LRCLK
NC			HPWR	25	26	IIS_EN	GPIO34	GPIO34	PDM_DAT
NC			HPWR	27	28	5V			5V
NC			HPWR	29	30	BATTERY			BAT

GPIOXX same
GPIOXX M5Stack only
GPIOXX M5Core2 only
GPIOXX moved

Flash 1)	
GPIO6	SCLK
GPIO7	SO
GPIO8	SI
GPIO9	HOLD#
GPIO10	WP#
GPIO11	CS#

1) M5Stack and M5Core2

PSRAM 2)	
GPIO17	SCLK
GPIO7	SO/SIO1
GPIO8	SI/SIO0
GPIO9	SIO3
GPIO10	SIO2
GPIO16	CS#

2) Also present in M5Stack Fire

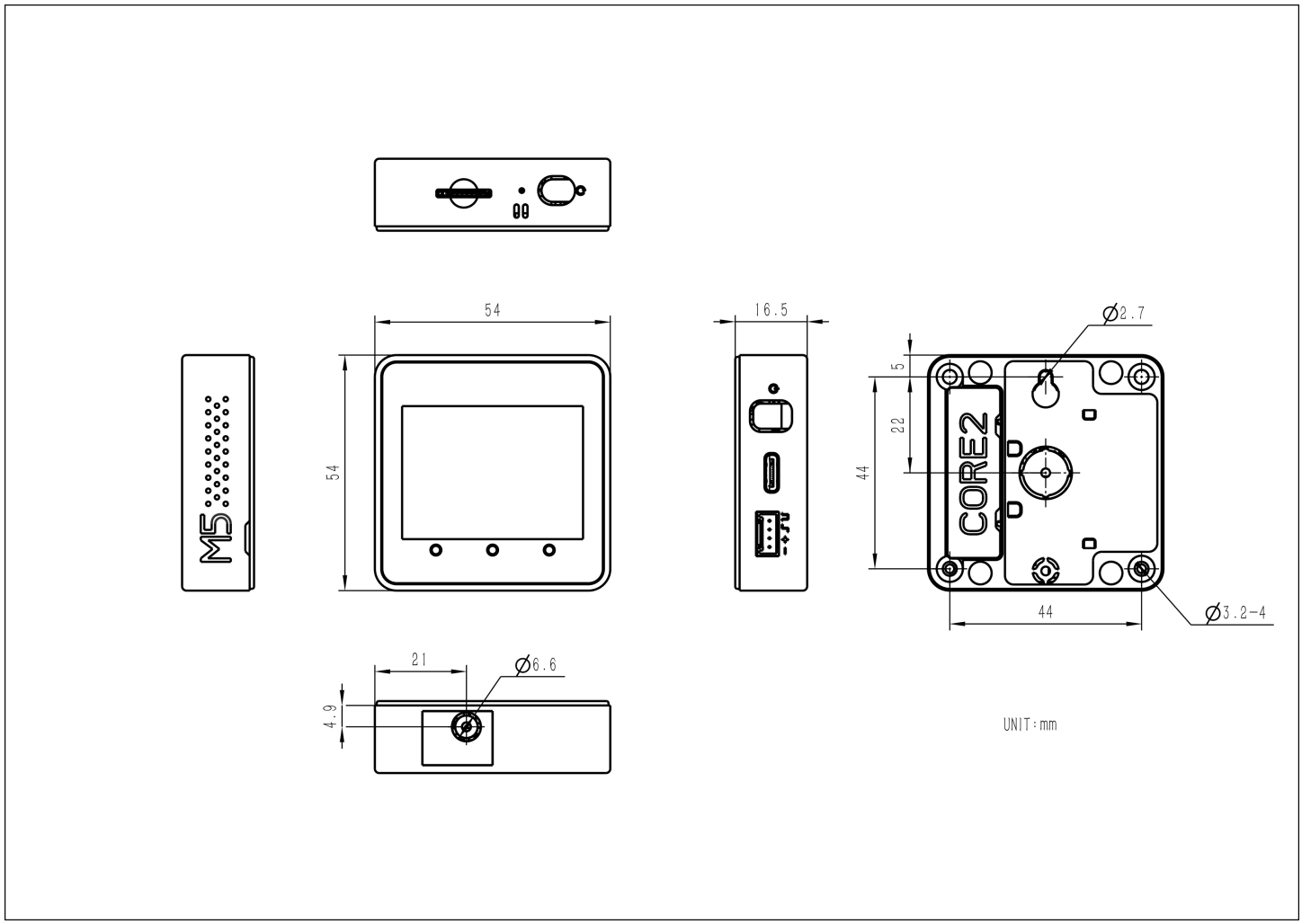
M5Core2	
LCD	
GPIO23	MOSI
GPIO38	MISO
GPIO18	SCK
GPIO5	CS
GPIO15	D/C
AXP_IO4	Reset
AXP_DCDC3	Backlight
Touch	
GPIO21	SDA
GPIO22	SCK
GPIO39	INT
SDCard	
GPIO23	CMD/MOSI
GPIO38	DAT0/MISO
GPIO18	CLK/SCK
GPIO4	DAT3/CS
Speaker	
GPIO0	LRCK
GPIO12	BCLK
GPIO2	SADTA
AXP_IO2	Enable
RTC/AXP	
GPIO21	SDA
GPIO22	SCK
SYS_LED	
AXP_IO1	on

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M5Core2 Port Description

PORT	PIN	Note:
PORT-A (Red)	G32/33	I2C
PORT-B (Black)	G26/36	DAC/ADC
PORT-C (Blue)	G13/14	UART

Model Size



Structure

- Core2 Structure Files

Datasheets

- ESP32
- FT6336U
- NS4168
- MPU6886
- ILI9342C
- SPM1423
- BM8563
- SY7088

- AXP192 datasheet
- AXP192 register
- 1027DC Motor
- ILI9342E

Softwares

Arduino

- Core2 Arduino Quick Start
- Core2 Arduino Driver Library
- Core2 Arduino API

UiFlow1

- Core2 UiFlow1 Quick Start

UiFlow2

- Core2 UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-core2]
platform = espressif32@6.12.0
board = m5stack-core2
framework = arduino
upload_speed = 921600
monitor_speed = 115200
board_build.partitions = default_16MB.csv
build_type = debug
build_flags =
    -DBOARD_HAS_PSRAM
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

ESP-IDF

- Core2 ESP-IDF BSP Tutorial

USB Driver

Driver

Click the link below to download the driver for your operating system. There are currently two driver chip versions, CP210X (for **CP2104** version) / CP34X (for **CH9102** version) driver package. After extracting the package, select the installation package for your operating system's bit version to install. (If you are unsure which USB chip your device uses, you can install both drivers. **CH9102_VCP_SER_MacOS v1.7** may report an error during installation, but it has actually completed the installation, so you can ignore it.) If you encounter issues with downloading programs (such as timeout or "Failed to write to target RAM"), try reinstalling the device driver.

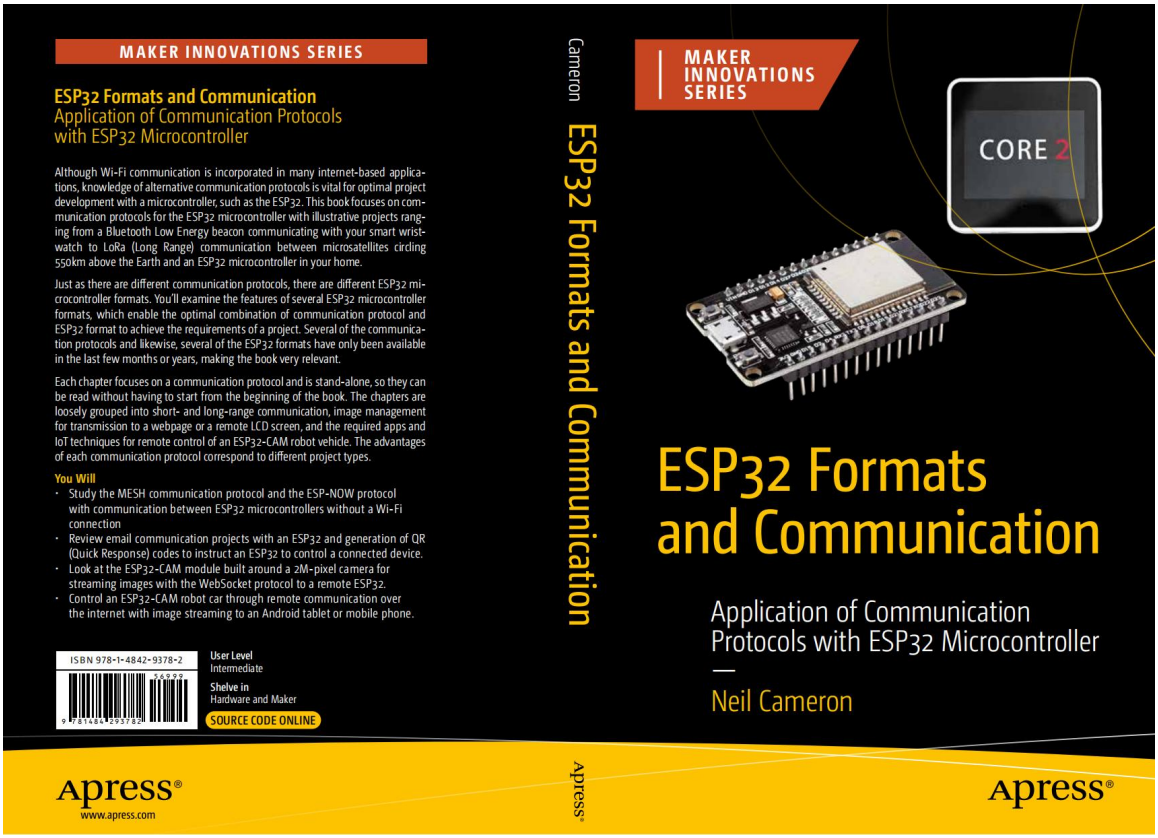
Driver Name	Applicable Driver Chip	Download Link
CP210x_VCP_Windows	CP2104	Download
CP210x_VCP_MacOS	CP2104	Download
CP210x_VCP_Linux	CP2104	Download
CH9102_VCP_SER_Windows	CH9102	Download
CH9102_VCP_SER_MacOS v1.7	CH9102	Download

Easyloader

Easyloader	Download Link	Note
Core2 Factory FirmWare	download	/

Other

- Core2 Factory Firmware Restore Tutorial
- ESP32 formats and communication protocols



The book 《ESP32 formats and communication protocols》 introduces the M5Stack Core2 module in several chapters. The M5Stack Core2 module integrates a touch LCD screen, and Wi-Fi communication, a microphone and speaker, as well as an accelerometer and gyroscope, making the M5Stack Core2 module very versatile. The book uses communication protocols to build projects, from connecting a smartwatch to a phone (BLE) to remote communication with satellites orbiting above the earth (LoRa) and audio signal transmission between devices (I2S). QR codes are used to control external devices via the internet, while ESP-MESH and ESP-NOW protocols enable communication between microcontrollers without an internet connection.

Video

Product Comparison

To compare information on the controller series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.

Version Change

Release Date	Product Changes	Note
2026.8.7	LCD screen model changed from ILI9342C to ILI9342E	When compiling programs with the M5GFX library, M5GFX library version >= 0.2.27 is required
2023.10	Lithium battery capacity changed from 390mAh to 500mAh	/
2023.2	Removed RTC button battery	Does not affect timing functions
2021.7	CP2104 replaced with CH9102F	Actual shipments include CP2104 and CH9102F chip versions; there is no difference in function or use
2020.6	First release	/